

VDA9.5

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Exercise 5 (Asymptotic Decider, 6+3 Bonus Points)

a) According to Chapter 9, the direct way to evaluate the asymptotic decider involves re-writing the standard equation for bilinear interpolation into a form that allows us to read off the intersection point of the asymptotes and the function value at that point. Specify explicit formulas to compute μ , x_0 , y_0 , and γ from $f_{i,j}$, $f_{i+1,j}$, $f_{i,j+1}$, and $f_{i+1,j+1}$. (4P)

b) In cases which require the asymptotic decider, do we ever have to divide by zero in the equations you derived in a)? Why? (2P)

c) Unfortunately, the scientific paper that introduced the asymptotic decider contained a serious typo. You can download this paper from eCampus. We will give you a bonus point if you identify the equation in that paper where the equivalent of γ (in our notation) is computed, and two more if you tell us what is wrong with this equation, and how to fix it.

a)

We remember from the lecture that the bilinear interpolation in terms of the corner values has the form $f(x, y) = (1 - x)(1 - y)f_{i,j} + x(1 - y)f_{i+1,j} + (1 - x)yf_{i,j+1} + xyf_{i+1,j+1}$. We have to transform this expression into the form $f(x, y) = \mu(x - x_0)(y - y_0) + \gamma$. We expand the terms:

$$\begin{aligned} f(x, y) &= (1 - x)(1 - y)f_{i,j} + x(1 - y)f_{i+1,j} + (1 - x)yf_{i,j+1} + xyf_{i+1,j+1} = \\ &= xy(f_{i,j} - f_{i+1,j} - f_{i,j+1} + f_{i+1,j+1}) - x(f_{i,j} - f_{i+1,j}) - y(f_{i,j} - f_{i,j+1}) + f_{i,j} \end{aligned}$$

By setting $\mu = f_{i,j} - f_{i+1,j} - f_{i,j+1} + f_{i+1,j+1}$:

$$\begin{aligned} f(x, y) &= \mu xy - \mu x \frac{f_{i,j} - f_{i+1,j}}{\mu} - \mu y \frac{f_{i,j} - f_{i,j+1}}{\mu} + \mu \frac{(f_{i,j} - f_{i+1,j})(f_{i,j} - f_{i,j+1})}{\mu^2} \\ &\quad - \mu \frac{(f_{i,j} - f_{i+1,j})(f_{i,j} - f_{i,j+1})}{\mu^2} + f_{i,j} = \\ &= \mu \left(x - \frac{f_{i,j} - f_{i,j+1}}{\mu} \right) \left(y - \frac{f_{i,j} - f_{i+1,j}}{\mu} \right) - \frac{(f_{i,j} - f_{i+1,j})(f_{i,j} - f_{i,j+1})}{\mu} + f_{i,j} \end{aligned}$$

Thus, we can set $x_0 = \frac{f_{i,j}-f_{i,j+1}}{\mu}$, $y_0 = \frac{f_{i,j}-f_{i+1,j}}{\mu}$, $\gamma = f_{i,j} - \frac{(f_{i,j}-f_{i+1,j})(f_{i,j}-f_{i,j+1})}{\mu}$, and we have that $f(x, y) = \mu(x - x_0)(y - y_0) + \gamma$. Thus, the asymptotes of the hyperbola are found at the lines $x = x_0$ and $y = y_0$, and the value there would be $f(x_0, y_0) = \gamma$, which we could compare directly with our isovalue to determine the case.

b)

We only need to use the asymptotic decider when $f_{i,j}, f_{i+1,j+1} > c$, $f_{i+1,j}, f_{i,j+1} \leq c$ or viceversa, for the isovalue c . But in this case we have that $\mu = f_{i,j} - f_{i+1,j} - f_{i,j+1} + f_{i+1,j+1} > c - f_{i+1,j} - f_{i,j+1} + c \geq 2c - c - c = 0$. Thus $\mu > 0$, and we wouldn't divide by zero. The case $f_{i,j}, f_{i+1,j+1} \leq c$, $f_{i+1,j}, f_{i,j+1} > c$ would similarly yield the result $\mu < 0$.

c)

The equations analogous to the previous one are found at page 3 (85 in the paper's page order). In particular the author uses the symbols $B_{00}, B_{01}, B_{10}, B_{11}$ for the values of the function at the corners, and the symbols S_α, T_α for x_0, y_0 respectively. The autor uses the term $B(S_\alpha, T_\alpha)$ as an indicator of the interpolation's value at the intersection of the asymptotes, which is what we determined to be γ . The author asserts that $B(S_\alpha, T_\alpha) = \frac{B_{00}B_{11}+B_{01}B_{10}}{B_{00}-B_{10}-B_{10}+B_{11}}$, which in our notation would imply that $\gamma = \frac{f_{i,j}f_{i+1,j+1}+f_{i+1,j}f_{i,j+1}}{\mu}$. However we can expand our expression for γ :

$$\begin{aligned} \gamma &= f_{i,j} - \frac{(f_{i,j} - f_{i+1,j})(f_{i,j} - f_{i,j+1})}{\mu} = \\ &= \frac{f_{i,j}f_{i,j} - f_{i,j}f_{i+1,j} - f_{i,j}f_{i,j+1} + f_{i,j}f_{i+1,j+1} - f_{i,j}f_{i,j} + f_{i,j}f_{i+1,j} + f_{i,j}f_{i,j+1} - f_{i+1,j}f_{i,j+1}}{\mu} = \\ &= \frac{f_{i,j}f_{i+1,j+1} - f_{i+1,j}f_{i,j+1}}{\mu} \end{aligned}$$

Which in the author's original notation would imply that $B(S_\alpha, T_\alpha) = \frac{B_{00}B_{11}-B_{01}B_{10}}{B_{00}-B_{10}-B_{10}+B_{11}}$, so he forgot the negative sign in the second term.